

Notes: Kepler, Naiker, and Stewar (JF, 2023)

Stealth Acquisitions and Product Market Competition

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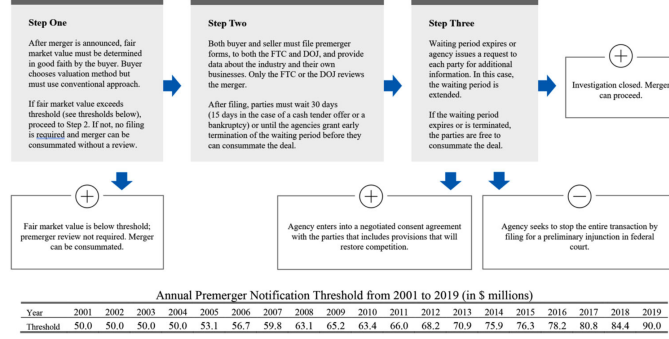


Figure 1. Pre-merger notification and review process. The Hart-Scott-Rodino Act established the federal pre-merger notification program, which provides the Federal Trade Commission and the Department of Justice with information about mergers and acquisitions before they become effective. This figure depicts the pre-merger notification and review process from start to completion. Arrows indicate the flow of the process. Positive symbols indicate that the merger closes successfully, whereas a negative symbol indicates that the agency seeks to prevent the merger from closing. (Color figure can be viewed at wileyonlinelibrary.com)

1 Overview

This paper studies *stealth acquisitions*: mergers and acquisitions (M&As) that appear to be deliberately structured to fall just below the U.S. Federal Trade Commission (FTC) pre-merger notification threshold under the Hart-Scott-Rodino (HSR) Act, thereby avoiding the standard pre-merger review process. The authors (i) document a sharp discontinuity in deal frequencies around the time-varying HSR size-of-transaction threshold, (ii) examine which deal features and contract terms are associated with being just below the threshold, and (iii) test whether these below-threshold (and plausibly more anticompetitive) acquisitions are followed by patterns consistent with reduced product-market competition, including positive announcement returns for horizontal rivals, increases in industry gross margins, and higher consumer prices in a retail-scanner setting.

2 Institutional background and why the threshold matters

2.1 HSR pre-merger notification as an economic “notch”

Under the HSR Act, transactions above a size-of-transaction threshold generally require a *pre-merger notification filing* and create a waiting period during which the FTC/DOJ can review the deal and potentially issue a *Second Request* (additional information) or seek to block/alter the merger. Transactions below the threshold are typically not subject to this standard pre-merger review (though post-merger challenges remain possible).

2.2 Figure 1: Process map

Figure 1 provides a flow-chart of the notification and review process. The key economic implication is that crossing the threshold can impose (i) direct compliance and delay costs and (ii) a higher probability of regulatory intervention. Therefore, if firms can adjust reported transaction value, they may have incentives to keep deals just below the threshold.

3 Data and descriptive statistics

3.1 Data sources (high level)

The core deal data come from SDC Mergers and Acquisitions, focusing on U.S. public and private M&As. The paper additionally uses (i) deal-contract features (e.g., earnouts, payment form, certain legal terms), (ii) stock-market data to form announcement-window returns for industry rivals, (iii) accounting data to compute changes in industry gross margins, and (iv) retail scanner data to examine product prices for “common products” in selected horizontal-merger settings.

3.2 Table I: Sample selection and industry distribution

Panel A of **Table I** shows the sample construction: starting from 34,839 M&As, the authors apply exclusions (missing ownership data, financials/utilities, real estate/hotels, subsidiary clean-ups) to reach a **full sample of 19,886 deals**.

Table I Sample Selection and M&A Descriptive Statistics		
This table presents the sample selection procedure for the full sample of M&As (Panel A) and for the near-threshold sample of M&As (Panel B). The sample is constructed using the universe of deals from the Securities Data Company (SDC) Mergers and Acquisition database announced between February 1, 2001 and February 27, 2020. In addition, Panel C presents the top 10 industries in our full sample of 19,886 deals , using Fama-French 48-industry classifications, Panel D presents the comparable distributions for the top 10 industries in two subsamples representing deals with transaction values that are within 10% of the annual FTC notification threshold, that is deal values that are either $\geq 0\%$ but $\leq 10\%$ below the threshold or $> 0\%$ but $\leq 10\%$ above the threshold (i.e., just-below-threshold deals and just-above-threshold deals, respectively). Panel B presents the top 10 industries separately for below- and above-threshold deals.		
Panel A: Sample Selection (Full Sample)		
All U.S. public and private M&As ($> \$1$ mil.) from February 1, 2001 to February 6, 2020		34,839
Less: Deals with missing data on % acquired, % owned before, or % owned after		(6,225)
Less: Deals involving financial firms or utilities		(7,212)
Less: Deals involving real property for rental or investment purposes, or hotels		(1,495)
Less: Deals when acquirer purchases remaining interest of its own subsidiary		(21)
Full sample of deals		19,886
Panel B: Near-Threshold Sample		
Just-above-threshold M&As		274
Just-below-threshold M&As		366
Sample of near-threshold deals		640
Panel C: Industry Distribution (Top Ten Industries)		
Fama-French 48-Industry Groups	Number of Deals	Percentage of All Deals
Top 10		
Business services	6,417	32.27
Pharmaceutical products	1,305	6.56
Healthcare	1,012	5.09
Electronic equipment	991	4.98
Wholesale	776	3.90
Retail	753	3.79
Medical equipment	716	3.60
Communication	665	3.34
Computers	570	2.87
Transportation	503	2.53
Total (Top 10)	13,708	66.40

Table I—Continued				
Panel D: Industry Distribution (Within $\pm 10\%$ of Annual Threshold)				
Fama-French 48-Industry Groups	Just-Below Threshold		Just-Above Threshold	
	Number of Deals	Percentage of Deals “Below”	Number of Deals	Percentage of Deals “Above”
Top 10				
Business services	123	33.61	93	33.94
Pharmaceutical products	22	6.01	17	6.20
Electronic equipment	22	6.01	11	4.01
Medical equipment	20	5.46	13	4.74
Retail	18	4.92	15	5.47
Healthcare	13	3.55	16	5.84
Wholesale	13	3.55	7	2.55
Personal services	12	3.28	7	2.55
Computers	12	3.28	13	4.74
Construction	10	2.73	3	1.09
Total (Top 10)	265	72.40	195	71.13

Panel B defines the *near-threshold* sample within $\pm 10\%$ of the annual threshold: **366 just-below** and **274 just-above** deals, for a total of **640 near-threshold deals**.

Panels C–D show the industry distribution (Fama–French 48). Business services dominates both the full sample and near-threshold subsamples, and the near-threshold industry mix is broadly similar to the full-sample mix. This matters for interpretation because it reduces concern that one “special” industry mechanically drives the near-threshold patterns.

Table II
Descriptive Statistics

This table presents the distribution of key variables used in our analysis. All variables are defined in Appendix A. Panel A presents descriptive statistics for all variables for both the pooled and the near-threshold analysis, and Panel B presents descriptive statistics for key variables in the near-threshold analysis (split by below versus above the pre-merger review threshold). *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Key Variables (Pooled and Near-Threshold Analysis)

Variable	N	Mean	SD	25 th	Median	75 th
<i>Deal analysis</i>						
PublicAcquirer	19,886	0.69	0.46	0.00	1.00	1.00
PrivateAcquirer	19,886	0.15	0.36	0.00	0.00	0.00
PublicTarget	19,886	0.23	0.42	0.00	0.00	0.00
Public-Public	19,886	0.12	0.33	0.00	0.00	0.00
Public-Private	19,886	0.57	0.49	0.00	1.00	1.00
Private-Private	19,886	0.09	0.29	0.00	0.00	0.00
Private-Public	19,886	0.06	0.24	0.00	0.00	0.00
AllCash	19,886	0.32	0.47	0.00	0.00	1.00
AllStock	19,886	0.08	0.28	0.00	0.00	0.00
AllCashandOther	19,886	0.78	0.41	1.00	1.00	1.00
AllStockandOther	19,886	0.48	0.50	0.00	1.00	1.00
Horizontal	19,886	0.28	0.45	0.00	0.00	1.00
Intrastate	19,886	0.19	0.39	0.00	0.00	0.00
Earnouts	19,886	0.10	0.30	0.00	0.00	0.00
EarnoutPct	19,886	3.60	13.11	0.00	0.00	0.00
AcqTermFeePercent	19,886	0.00	0.02	0.00	0.00	0.00
PublicTargetDealPremium	5,642	64.58	456.29	14.34	31.46	54.28
EconomicTie	603	0.75	0.43	0.00	1.00	1.00
PrivateTargetDealPremium	222	55.74	22.84	40.63	57.21	73.40
EarnoutPayoff	45	0.69	0.47	0.00	1.00	1.00
<i>Financial contracting analysis</i>						
ExtendedLiabilityCoverage	234	0.53	0.50	0.00	1.00	1.00
DeductibleThreshold	192	0.33	0.47	0.00	0.19	0.50
<i>Returns and gross margin analysis</i>						
RivalRet	594	0.00	0.03	-0.01	0.00	0.01
ΔGrossMargin	364	0.01	0.04	-0.01	0.00	0.02
<i>Product pricing analysis</i>						
NormalizedPrice	1,915,291	0.05	1.02	-0.70	0.00	0.71
JustBelowThreshold	1,915,291	0.21	0.41	0	0	0
Post	1,915,291	0.49	0.50	0	0	1
<i>Product pricing analysis</i>						
<i>Below-Threshold</i>						
Price	77,835,861	6.46	3.31	3.99	5.67	8.09
CommonProduct	77,835,861	0.01	0.07	0.00	0.00	0.00
Post	77,835,861	0.46	0.50	0.00	0.00	1.00
<i>Above-Threshold</i>						
Price	4,091,337	19.65	25.16	6.79	9.99	23.74
CommonProduct	4,091,337	0.06	0.24	0.00	0.00	0.00
Post	4,091,337	0.49	0.50	0.00	0.00	0.00
<i>Further-Below-Threshold</i>						
Price	4,760,492	3.64	0.91	3.00	3.69	3.99
CommonProduct	4,760,492	0.26	0.44	0.00	0.00	1.00
Post	4,760,492	0.53	0.50	0.00	1.00	1.00
<i>Controls</i>						
DealValue	19,886	400.33	2,625.67	7.50	26.00	120.00
TenderOffer	19,886	0.04	0.19	0.00	0.00	0.00
PrivateTarget	19,886	0.77	0.42	1.00	1.00	1.00
NumRivals	594	40.54	55.77	4.00	14.00	49.00
RepsSurvive	232	0.59	0.49	0.00	1.00	1.00
SurvivalPeriod	222	0.85	0.95	0.00	1.00	1.50
TargetTermFeePercent	19,886	0.01	0.03	0.00	0.00	0.00

Table II—Continued

Panel B: Near-Threshold Sample (±10% of Threshold)								
Variable	Just-Below Threshold			Just-Above Threshold			Diff in Means	Diff in Medians
	N	Mean	Median	N	Mean	Median		
<i>Deal analysis</i>								
PublicAcquirer	366	0.73	1.00	274	0.73	1.00	0.00	0.00
PrivateAcquirer	366	0.10	0.00	274	0.13	0.00	-0.03	0.00
PublicTarget	366	0.19	0.00	274	0.23	0.00	-0.04	0.00
Public-Public	366	0.08	0.00	274	0.12	0.00	-0.04*	0.00*
Public-Private	366	0.65	1.00	274	0.61	1.00	0.04	0.00
Private-Private	366	0.05	0.00	274	0.07	0.00	-0.02	0.00
Private-Public	366	0.05	0.00	274	0.07	0.00	-0.02	0.00
AllCash	366	0.36	0.00	274	0.33	0.00	0.02	0.00
AllStock	366	0.04	0.00	274	0.09	0.00	-0.05**	0.00***
AllCashandOther	366	0.85	1.00	274	0.76	1.00	0.09**	0.00***
AllStockandOther	366	0.42	0.00	274	0.47	0.00	-0.05	0.00
Horizontal	366	0.28	0.00	274	0.26	0.00	0.02	0.00
Horizontal(Continuous)	366	0.43	0.33	274	0.35	0.15	0.08**	0.18**
Intrastate	366	0.17	0.00	274	0.20	0.00	-0.03	0.00
Earnouts	366	0.16	0.00	274	0.11	0.00	0.05*	0.00*
EarnoutPct	366	4.62	0.00	274	3.57	0.00	1.05	0.00
AcqTermFeePercent	366	0.00	0.00	274	0.00	0.00	0.00**	-0.00**
PublicTargetDealPremium	50	90.46	47.36	51	65.62	49.05	27.84	-1.69
PrivateTargetDealPrem	133	55.00	56.60	89	56.87	56.60	-1.87	0.00
EconomicTie	343	0.80	1.00	260	0.68	1.00	0.12***	0.00***
EarnoutPayoff	28	0.64	1.00	17	0.76	1.00	-0.12	0.00
<i>Financial contracting analysis</i>								
ExtendedLiabilityCoverage	123	0.54	1.00	111	0.51	1.00	0.03	0.00
DeductibleThreshold	105	0.39	0.24	87	0.27	0.06	0.12**	0.18
<i>Returns and gross margin analysis</i>								
RivalRet	344	0.00	0.00	250	0.00	0.00	0.00	0.00
ΔGrossMargin	206	0.01	0.01	158	0.01	0.02	0.00	-0.01
<i>Controls</i>								
DealValue	366	59.35	58.27	274	66.66	65.13	-7.31***	-7.86***
TenderOffer	366	0.03	0.00	274	0.03	0.00	0.00	0.00
RepsSurvive	122	0.61	1.00	110	0.56	1.00	0.05	0.00
SurvivalPeriod	116	0.87	1.00	106	0.84	1.00	0.03	0.00
TargetTermFeePct	366	0.00	0.00	274	0.01	0.00	0.01	0.00*
NumRivals	344	39.66	14.00	250	41.76	13.50	-2.10	0.50

3.3 Table II: Descriptive statistics (pooled and near-threshold)

Table II, Panel A summarizes key variables. A few takeaways for orientation:

- The typical deal involves a *public acquirer* (mean ≈ 0.69 in the pooled sample).
- The authors work with multiple analysis samples (deal-level contracting, rival returns, gross margins, and product-level prices). Table II clarifies the sample sizes and the scale of each data layer (e.g., the product-price panel has millions of weekly observations).

Table II, Panel B compares just-below vs just-above deals within the $\pm 10\%$ window. Many observables look similar, but several economically relevant differences already appear:

- All-stock deals are *less* common just below the threshold, while “cash and other” structures are *more* common.
- Earnouts and CEO economic ties appear more prevalent among just-below deals.
- A “continuous” measure of horizontal relatedness is higher just below the threshold.

These patterns foreshadow the regression evidence in Tables IV–VIII.

4 Existence of stealth acquisitions: density discontinuity evidence

4.1 Key construction: standardizing the time-varying threshold

Let i index deals and t index years. The FTC threshold changes annually, so the authors standardize deal values by their signed distance from the threshold:

$$\text{DistanceFromThreshold}_{i,t} = \text{DealValue}_{i,t} - \text{Threshold}_t. \quad (1)$$

This converts a time-varying threshold into a single cutoff at zero.

4.2 McCrary density test intuition (Figure 2 and Table III, Panel A)

The core question is whether the density of deals is continuous at zero. Absent manipulation, there should be no discrete jump in the number of deals just below vs just above the cutoff. **Figure 2** plots kernel density estimates around zero and visually shows a sharp drop moving from just below to just above the threshold. **Table III, Panel A** formalizes this with the McCrary (2008)

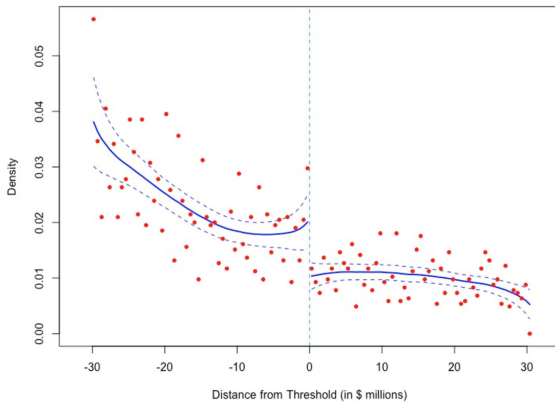


Figure 2. Density of public and private M&As around FTC pre-merger review threshold. This figure depicts the estimated kernel densities of the distance (in \$ millions) of deal values from the FTC pre-merger review threshold. Confidence bands (at the 95% level) are depicted by the thin lines. The circles, which describe scaled frequencies, are analogous to histogram bins. For ease of interpretation, we restrict our kernel density estimations to distances within $\pm \$30$ million of zero, where zero represents the FTC threshold. The estimation procedure follows the method in McCrary (2008). (Color figure can be viewed at wileyonlinelibrary.com)

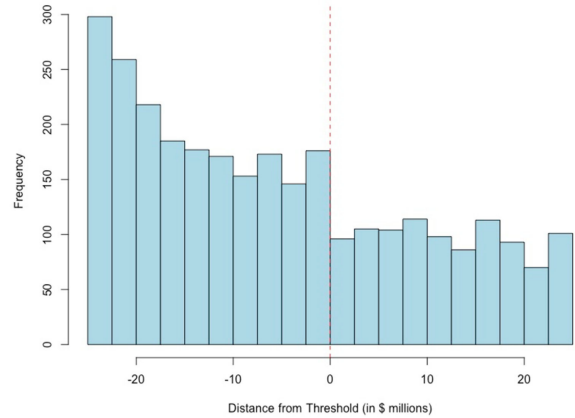


Figure 3. Histogram of M&A deals around FTC pre-merger review threshold. This figure depicts the frequency of deals involving public and private targets around the FTC pre-merger review threshold. For ease of interpretation, we restrict our histogram to deals within $\pm \$25$ million of zero, where zero represents the FTC threshold. Bin widths are set to \$2.5 million and constructed such that when the distance from the threshold equals zero, the deal is included in the bin to the left of the threshold (i.e., $[-2.5, 0)$), indicating the deal would be exempt from pre-merger review, which is consistent with the regulation. (Color figure can be viewed at wileyonlinelibrary.com)

density-discontinuity test: the reported *log difference in density heights* at the cutoff is strongly negative and statistically significant for the main sample, consistent with *excess mass* just below the threshold (i.e., stealth acquisition bunching).

4.3 Histogram bin test (Figure 3 and Table III, Panel B)

Figure 3 shows a histogram of deal counts around the cutoff. **Table III, Panel B** compares the *actual* counts in the bins immediately adjacent to the cutoff with *expected* counts based on adjacent-bin averages (Burgstahler and Dichev style). The immediate-left bin (just below) is substantially higher than expected, while the immediate-right bin (just above) is substantially lower than expected, reinforcing the density-discontinuity conclusion.

Table III
**Tests of the Difference between the Frequency of Below-Threshold
and Above-Threshold Deals**

This table presents results of tests of the statistical difference between the frequency of just-below-threshold deals and just-above-threshold deals. In Panel A, we report results for the difference in density heights around the threshold related to Figures 2 and 5. The log difference in heights is from the perspective of the bin just to the right of the threshold (i.e., a negative sign indicates the right bin is lower than the left bin). In Panel B, we report results for the difference between actual and estimated frequencies of deals occurring in the bins just to the left and just to the right of zero related to Figure 3. *** indicates significance at the 1% level. The estimation procedure follows the methods in McCrary (2008) in Panel A and Burgstahler and Dichev (1997) in Panel B.

Panel A: Difference in Density Heights				
	Log Diff. in Heights			t-Statistic
Difference around threshold (Figure 2)	-0.680***			-5.019
Falsification test (Figure 4)	0.129			0.445
Panel B: Difference in Estimated and Actual Bin Heights (Figure 3)				
Bin	Frequency (Actual)	Frequency (Estimated)	Difference	t-Statistic
JustBelowThreshold	176	121	55***	4.046
JustAboveThreshold	96	141	-45***	-3.103

4.4 Falsification tests (Figures 4–6 and Table III)

The paper then asks whether the discontinuity is unique to the true cutoff:

- **Figure 4** uses placebo thresholds (shifted cutoffs) and shows that the true cutoff yields the most extreme discontinuity statistic.
- **Figure 5** repeats the density test for deal types that are legally exempt from pre-merger review (real estate/hotels) and finds no discontinuity.
- **Figure 6** uses the *one-year-ahead* threshold (a wrong cutoff) and again finds no discontinuity.

Together, these support interpretation of the jump as active manipulation/avoidance rather than a generic feature of deal-size distributions.

5 Characteristics of stealth acquisitions: contracting and deal design

The remainder of Section III studies *how* deals just below the cutoff differ and what mechanisms may enable “stealth.”

5.1 Baseline linear probability model for being just-below (Table IV and related)

The paper uses an OLS linear probability model (LPM) with fixed effects:

$$\text{JustBelowThreshold}_{i,t} = \alpha + \beta X_{i,t} + \theta \text{Controls}_{i,t} + \psi_j + \delta_t + \varepsilon_{i,t}, \quad (2)$$

where $\text{JustBelowThreshold}_{i,t} = 1$ if deal value is within [0%, 10%] below the threshold and 0 otherwise; ψ_j are industry fixed effects and δ_t are year fixed effects. The authors emphasize OLS partly because multiple fixed effects make nonlinear binary models vulnerable to the incidental-parameters problem.

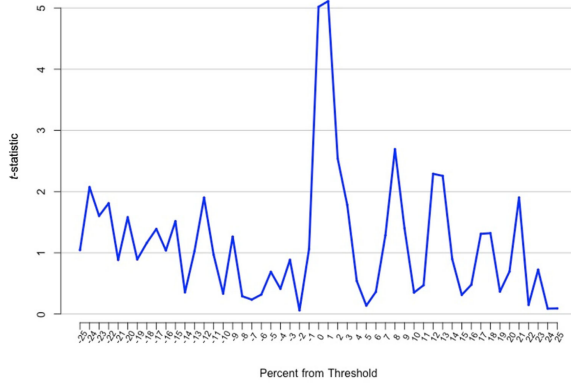


Figure 4. Falsification test: Placebo thresholds. This figure depicts t -statistics for discontinuities at points along the distribution of deals. The level of the t -statistic is measured on the y-axis. Thresholds, as the percentage distance from the actual threshold, are shown on the x-axis. The actual threshold is at zero. All other “placebo” thresholds begin at a distance of $\pm 1\%$ from the actual threshold and end at a distance of $\pm 25\%$ from the actual threshold. To construct a placebo threshold, we adjust the actual threshold in each year by $x\%$, where x is the same for each year, and then standardize all 19 years of “adjusted” thresholds. As an example, to measure the discontinuity when the placebo threshold is $+1\%$ from the actual threshold, we first adjust all thresholds (from 2001 to 2019) by 1%. We then standardize these new thresholds (at zero) and calculate the McCrary (2008) t -statistic at the discontinuity. For ease of interpretation, we display the absolute value of the t -statistic. (Color figure can be viewed at wileyonlinelibrary.com)

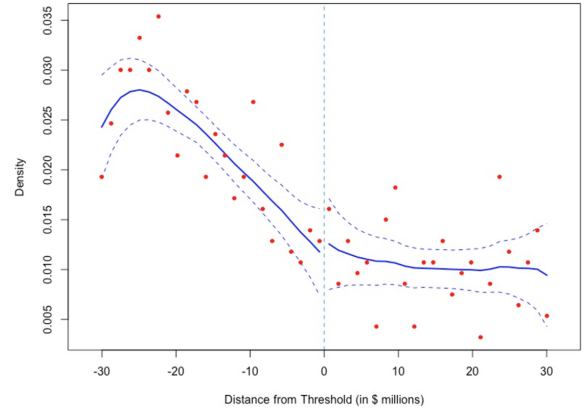


Figure 5. Falsification test: Density of always-exempt M&As involving public and private targets. This figure depicts the estimated kernel densities of the distance (in \$ millions) of deal values from the FTC pre-merger review threshold for deals involving real estate and hotels, which are always exempt from the pre-merger notification and review process. Confidence bands (at the 95% level) are depicted by the thin lines. The circles, which describe scaled frequencies, are analogous to histogram bins. For ease of interpretation, we restrict our kernel density estimations to distances within $\pm \$30$ million of zero, where zero represents the FTC threshold. The estimation procedure follows the method in McCrary (2008). (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Table IV: Acquirer–target type and payment form

Columns (1)–(4) show that some acquirer–target ownership pairings are systematically associated with just-below deals. In particular, *public acquirer buying a private target* is positively related to being just below the threshold, while *public–public* and *private–private* pairings are negatively related. Columns (5)–(8) show strong patterns in payment form:

Table IV
Acquirer-Target Characteristics and Below-Threshold M&As

This table presents results from OLS regressions of M&As on acquirer-target characteristics. The dependent variable, *JustBelow*, is an indicator that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. The main variables of interest in columns (1) to (4) are indicator variables that assume the value of 1 based on combined acquirer-target characteristics, and 0 otherwise. The main variables of interest in columns (5) to (8) are indicator variables that take the value of one if the deal's payment terms are structured as 100% cash, 100% stock, 100% cash and other (i.e., debt and earnouts), or 100% stock and other, respectively. We control for the size of the deal (*DealValue*) in all specifications and for public targets (*PublicTarget*) in columns (5) to (8). All variables are defined in Appendix A. All columns include target-firm industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>	(3) <i>JustBelow</i>	(4) <i>JustBelow</i>	(5) <i>JustBelow</i>	(6) <i>JustBelow</i>	(7) <i>JustBelow</i>	(8) <i>JustBelow</i>
Public-Public	-0.007** (-2.64)							
Public-Private		0.005** (2.71)						
Private-Private			-0.010*** (-3.40)					
Private-Public				-0.004 (-1.87)				
AllCash					0.005* (2.01)			
AllStock						-0.013*** (-12.16)		
AllCashandOther							0.008*** (4.01)	
AllStockandOther								-0.006* (-1.95)

- All-cash and “cash plus other” structures are more likely to be just below.
- All-stock and “stock plus other” structures are less likely to be just below.

A natural interpretation is that cash-heavy (and possibly more privately negotiated) deal structures may make it easier to adjust reported value or embed contingent components, whereas stock-heavy deals may be more transparent or harder to “fine-tune” around a cutoff.

5.3 Table V: Earnouts as a key avoidance technique

Earnouts are a canonical mechanism to keep *up-front* consideration below a threshold while preserving higher expected total consideration via contingencies. In **Table V**, the indicator for an earnout

Table IV—Continued

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>	(3) <i>JustBelow</i>	(4) <i>JustBelow</i>	(5) <i>JustBelow</i>	(6) <i>JustBelow</i>	(7) <i>JustBelow</i>	(8) <i>JustBelow</i>
DealValue	-0.000*** (-3.82)	-0.000*** (-5.96)	-0.000*** (-5.08)	-0.000*** (-5.80)	-0.000*** (-4.22)	-0.000*** (-4.49)	-0.000*** (-3.47)	-0.000*** (-4.47)
Constant	0.019*** (231.61)	0.016*** (17.09)	0.020*** (238.40)	0.019*** (362.25)	0.021*** (51.04)	0.013*** (132.57)	0.013*** (11.17)	0.023*** (18.50)
Observations	19,886	19,886	19,886	19,886	19,886	19,886	19,886	19,886
Adjusted R^2	0.001	0.002	0.002	0.001	0.002	0.002	0.001	0.002
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

provision is positively associated with being just below the threshold in the near-threshold sample. Conditional on having earnouts, the earnout percentage itself is not strongly related, suggesting that the *presence* of the contingency matters more than the exact fraction in this specification.

Table V
Financial Characteristics and Below-Threshold M&As

This table presents results from OLS regressions of M&As on financial contract characteristics. The sample is restricted to observations for which the deal value falls within a $\pm 10\%$ window centered on the FTC threshold (see Table I, Panel B). The dependent variable, *JustBelowThreshold*, is an indicator variable that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. The main variable of interest in column (1) is an indicator variable that assumes the value of 1 if the financial contract includes a provision for earnouts, and 0 otherwise. Results presented in column (2) are conditional on the inclusion of an earnout provision, where the main variable of interest is a continuous variable that measures the percentage of the transaction value represented by earnouts. All variables are defined in Appendix A. All columns include target-firm industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. **, *** indicate significance at the 5% and 1% level, respectively. The sample comprises 637 deals (base sample of 640 less three singletons).

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>
Earnouts	0.062** (2.42)	
EarnoutPerc		0.001 (0.66)
DealValue	-0.071*** (-11.01)	-0.108*** (-6.53)
Constant	5.022*** (12.60)	7.295*** (7.25)
Observations	637	79
Adjusted R^2	0.440	0.530
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

Table VI
Merger Agreement Terms and Below-Threshold M&As

This table presents results from OLS regressions of M&As on financial contract terms. The dependent variable, *JustBelowThreshold*, is an indicator that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. The main variable of interest in column (1), *ExtendedLiabilityCoverage*, is an indicator variable that assumes the value of 1 if the acquirer extends D&O coverage for the former directors and officers of the target, and 0 otherwise. The main variable of interest in column (2), *DeductibleThreshold*, is a continuous variable that measures the threshold (as a percentage of the total deal value) above which the target is responsible for post-acquisition claims against the acquirer. We also include controls for whether the representations and warranties survive beyond the effective date (*RepsSurvive*), the length of the survival period (*SurvivalPeriod*), and whether the holdback funds are held in escrow or otherwise (*Escrow*). All variables are defined in Appendix A. All columns include target-firm industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. **, *** indicate significance at the 5% and 1% levels, respectively. The sample comprises 122 deals in column (1) (base sample of 640 less 132 public targets less 366 deals with missing data to construct main independent variables less 12 deals with missing data to construct control variables less eight singletons). The sample comprises 99 deals in column (2) (base sample of 640 less 132 public targets less 379 deals for which terms do not survive beyond the closing date less 11 deals with missing data to construct main independent variable less nine deals with missing data to construct control variables less 10 singletons).

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>
ExtendedLiabilityCoverage	0.131** (2.16)	
DeductibleThreshold		18.705** (2.19)
DealValue	-0.108*** (-19.99)	-0.095*** (-14.36)
Constant	7.125*** (19.16)	6.020*** (11.56)
Observations	122	99
Adjusted R^2	0.603	0.609
Controls	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

5.4 Table VI: Merger agreement terms (D&O coverage and deductibles)

Table VI examines specific legal/insurance terms:

- **Extended liability (D&O) coverage** for target directors/officers is positively related to just-below deals.
- **Deductible thresholds** (how much loss the acquirer bears before claims shift to the target) are also positively related.

These terms can be interpreted as part of a bargaining package that compensates target insiders for accepting a lower stated deal value (to remain below the threshold) by improving their protection/expected payoffs on non-price dimensions.

5.5 Table VII: Deal premiums

Table VII studies whether just-below deals involve systematically different deal premiums. For private targets, the estimated coefficient is negative (and marginally significant), suggesting that just-below deals may feature lower measured premiums—consistent with the idea that targets accept lower reported prices in exchange for other forms of compensation (contingencies, insurance protections, employment/retention, etc.). For public targets, the premium coefficient is not meaningfully different.

Table VII
Private-Target Deal Premiums and Below-Threshold M&As

This table presents results from OLS regressions of M&As on private-target deal premiums. The dependent variable, *JustBelowThreshold*, is an indicator that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. The main variable of interest in column (1), *PrivateTargetDealPrem*, is a continuous variable that measures the premium paid for private targets. The main variable of interest in column (2), *PublicTargetDealPrem*, is a continuous variable that measures the premium acquirers pay for publicly traded target firms. All variables are defined in Appendix A. All columns include target-firm industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample comprises 219 deals in column (1) (base sample of 640 less 132 public targets less 102 deals with nonpublic acquirers less 184 deals with missing data to construct main independent variable less three singletons). The sample comprises 84 deals in column (2) (base sample of 640 less 508 private targets less 31 deals with missing data to construct main independent variable less 17 singletons).

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>
PrivateTargetDealPrem	-0.199* (-1.96)	
PublicTargetDealPrem		-0.000 (-0.25)
DealValue	-0.069*** (-7.54)	-0.060** (-2.80)
Constant	4.996*** (9.57)	4.030*** (3.19)
Observations	219	84
Adjusted <i>R</i> ²	0.449	0.373
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

Table VIII

Acquirer-Target CEO Economic Ties and Below-Threshold M&As

This table presents results from OLS regressions of M&As on acquirer-target CEO economic ties. The dependent variable, *JustBelowThreshold*, is an indicator that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. In Panel A, the main variable of interest, *EconomicTie*, is an indicator that takes the value of one if the target CEO is retained by the acquiring firm and/or has an economic interest in the surviving firm. We also control for whether the acquirer is public (*PublicAcquirer*) and for whether the merger is horizontal (*Horizontal*). In Panel B, column (1), the main variable of interest is *EarnoutPayoff*, and in column (2) the main variable of interest in the interaction term *EconomicTie* × *EarnoutPayoff*. All variables are defined in Appendix A. We include target-firm industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. **, *** indicate significance at the 5% and 1% levels, respectively. The sample comprises 423 deals in Panel A (base sample of 640 less 208 deals with missing data on economic ties less nine singletons). The sample comprises 39 (29) deals in column (1) ((2)) of Panel B (base sample of 640 less 551 deals without earnouts less 44 deals with missing data on earnout payoffs less six singletons).

Panel A: Economic Ties and Below-Threshold M&As

Dependent Variable	(1) <i>JustBelow</i>
EconomicTie	0.084** (2.15)
DealValue	-0.073*** (-8.67)
Constant	5.341*** (9.66)
Observations	423
Adjusted <i>R</i> ²	0.466
Controls	Yes
Year fixed effects	Yes
Industry fixed effects	Yes

Panel B: Economic Ties and Earnout Payoffs

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>
EarnoutPayoff	0.059 (0.49)	-0.386 (-1.78)
EconomicTie		-0.198 (-1.31)
EconomicTie × EarnoutPayoff		0.590*** (9.04)
DealValue	-0.144*** (-10.12)	-0.151*** (-6.06)
Constant	9.190*** (12.11)	9.682*** (6.84)
Observations	39	29
Adjusted <i>R</i> ²	0.753	0.505
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

5.6 Table VIII: CEO economic ties and incentives

Table VIII, Panel A shows that target-CEO economic ties (retention and/or continuing economic interest) are positively associated with being just below the threshold. This fits an incentives story: target managers may prefer avoidance if it increases closing probability or reduces regulatory delay, especially when they personally benefit from post-merger outcomes.

Panel B (small sample) suggests that earnout payoff sensitivity is especially pronounced when combined with CEO economic ties (a strong positive interaction term), consistent with a joint mechanism in which (i) contingent payments plus (ii) CEO continuing stakes jointly support “stealth” structuring.

5.7 Interactions and heterogeneity: coordination incentives (Table IX)

To study whether the just-below propensity is concentrated in settings where anticompetitive motives are more plausible, the paper estimates an interaction LPM:

$$\text{JustBelowThreshold}_{i,t} = \alpha + \beta(Z_{i,t} \times W_{i,t}) + Z_{i,t} + W_{i,t} + \theta \text{Controls}_{i,t} + \psi_j + \delta_t + \varepsilon_{i,t}, \quad (3)$$

where *Z* and *W* capture characteristics such as *horizontal* overlap, *intrastate* operations, and *high industry concentration*. Table IX, Panel A shows:

- Horizontal deals are more likely to be just below the threshold.

Table IX
Acquirer-Target Industry and Location and Below-Threshold M&As

This table presents results from OLS regressions of M&As on acquirer-target industry and location characteristics. The sample is restricted to observations for which the deal value falls within a $\pm 10\%$ window centered around the FTC threshold. In Panels A and B, the dependent variable, *JustBelowThreshold*, is an indicator variable that assumes the value of 1 if a deal's transaction value is within the 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. In Panel A, the main variables of interest in columns (1) and (2) are indicator variables that assume the value of one based on whether the target and acquirer share the same four-digit SIC code (i.e., horizontal merger) or share the same state of operations (i.e., intrastate). In column (3), the main variable of interest is the interaction term, *Horizontal* $\times$ *Intrastate*, which takes the value of 1 if the merger is both horizontal and intrastate, and 0 otherwise. All variables are defined in Appendix A. In Panel B, columns (1) and (4), the main variables of interest, *HighConc*, is an indicator that assumes the value of 1 if the target firm's industry is above the median concentration, and 0 otherwise. In columns (2) and (5) and (6), the main variables of interest are interaction terms *Horizontal* $\times$ *HighConc* and *Intrastate* $\times$ *HighConc*, which assume the value of 1 when the target firm's industry is above the median concentration and the acquirer and target share the same four-digit SIC code (in columns (2) and (5)), or share the same state of operations (in columns (3) and (6)), and 0 otherwise. In columns (1) to (3) industry concentration is estimated using the methodology in Hoberg and Phillips (2010) (*Conc_HP*). In columns (4) to (6), industry concentration is estimated using net sales by four-digit SIC code, by year (*Conc_Sales*) (Hsu and Robinson (2006)). All variables are defined in Appendix A. All columns in Panel A include target-firm industry fixed effects (using Fama-French 48-industry classifications) and year fixed effects, while all columns in Panel B include industry fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the target-firm industry and year levels. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample comprises 637 deals in Panel A (base sample of 640 less three singletons). The sample comprises 501 in columns (4) to (6) in Panel B (base sample of 640 less 136 deals with missing data to construct HHI measure less three singletons).

Panel A: Horizontal and Intrastate M&As

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>	(3) <i>JustBelow</i>
Horizontal	0.066** (2.61)		0.028 (1.43)
Intrastate		-0.061 (-1.31)	-1.28* (-1.92)
Horizontal $\times$ Intrastate			0.194 (2.91)
DealValue	-0.072*** (-10.81)	-0.072*** (-11.06)	-0.072*** (-11.45)

Panel B: Highly Concentrated Industry M&As

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>	(3) <i>JustBelow</i>	(4) <i>JustBelow</i>	(5) <i>JustBelow</i>	(6) <i>JustBelow</i>
Constant	5.048*** (12.85)	5.064*** (12.60)	5.076*** (12.66)	5.076*** (12.66)	5.076*** (12.66)	5.076*** (12.66)
Observations	637	637	637	637	637	637
Adjusted R ²	0.442	0.440	0.447	0.447	0.447	0.447
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Highly Concentrated Industry M&As

Dependent Variable	(1) <i>JustBelow</i>	(2) <i>JustBelow</i>	(3) <i>JustBelow</i>	(4) <i>JustBelow</i>	(5) <i>JustBelow</i>	(6) <i>JustBelow</i>
Measure of <i>HighConc</i>						
<i>Conc_HP</i>	-0.003 (-1.12)	-0.024 (-1.15)	-0.009 (-0.36)	0.011 (0.28)	-0.014 (-0.27)	0.050 (1.30)
Horizontal						
Horizontal $\times$ <i>HighConc</i>						
Intrastate						
Intrastate $\times$ <i>HighConc</i>						
DealValue	-0.071*** (-11.11)	-0.072*** (-11.32)	-0.072*** (-11.47)	-0.074*** (-10.45)	-0.074*** (-10.64)	-0.075*** (-11.50)
Constant	5.035*** (12.53)	5.055*** (12.76)	5.069*** (13.01)	5.111*** (13.86)	5.110*** (13.10)	5.173*** (12.96)
Observations	640	640	640	640	640	640
Adjusted R ²	0.444	0.447	0.446	0.460	0.464	0.467
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	No	No	No

- Intrastate deals alone are not robustly more likely.

- The interaction **Horizontal** $\times$ **Intrastate** is strongly positive, indicating that *local horizontal mergers* are particularly prone to stealth structuring.

Table IX, Panel B shows that **Horizontal** $\times$ **High Concentration** is positive and significant (across two concentration measures), while high concentration alone is not. This implies that stealth behavior is especially associated with *horizontal mergers in already concentrated industries*—exactly the cases where antitrust scrutiny would be most salient.

6 Product-market consequences of stealth acquisitions

Section IV asks whether below-threshold avoidance is associated with outcomes consistent with *reduced competition*.

6.1 Rival announcement returns (Table X; Figure intuition)

The idea (Eckbo-style) is that in an anticompetitive horizontal merger, industry rivals can earn monopoly rents too (by “free riding” on higher prices). If markets are efficient, rivals’ stock prices should react positively at announcement.

The paper estimates:

$$y_{i,[1,1]} = \alpha + \beta_1 (\text{JustBelowThreshold}_{i,t} \times \text{Horizontal}_{i,t}) + \beta_2 \text{Horizontal}_{i,t} \quad (1)$$

$$+ \beta_3 \text{JustBelowThreshold}_{i,t} + \theta \text{Controls}_{i,t} + \psi_j + \delta_t + \varepsilon_{i,t}. \quad (4)$$

Table X
Horizontal Rivals’ Announcement Returns, Gross Margins, and Below-Threshold M&As

This table presents results from OLS regressions of announcement returns and gross margins on M&As. In columns (1) and (2), the dependent variable, *RivalRet*, is a continuous variable that represents the equal-weighted (three-day) market-adjusted portfolio returns of horizontal rivals of acquirers. In columns (3) and (4), the dependent variable, $\Delta \text{GrossMargin}$, is a continuous variable that equals the change in the industry-average gross margin measured as the difference between the industry-average gross margin before the merger and the industry-average gross margin after the merger, where gross margin equals sales minus cost of goods sold all scaled by sales. The main variable of interest in columns (1) and (3), *JustBelowThreshold*, is an indicator variable that assumes the value of 1 if a deal's transaction value is within a 10% window below the FTC annual pre-merger review threshold, and 0 otherwise. In columns (2) and (4), we present results for the association between announcement returns (in column (2)) and gross margins (in column (4)) and an interaction term, *Horizontal* $\times$ *JustBelowThreshold*, which assumes the value of 1 if the deal is below the threshold and the acquirer and target share the same four-digit SIC code, and 0 otherwise. All regressions also include as controls *DealValue*, *PublicAcquirer*, *PublicTarget*, and *LowNumRivals*. All variables are defined in Appendix A. All columns include acquirer industry fixed effects (using Fama-French 48-industry classification) and year fixed effects. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the acquirer industry and year levels. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The sample comprises 543 deals in columns (1) and (2) (base sample of 640 less 94 deals with missing data to construct variables less three singletons). The sample comprises 359 deals in columns (3) and (4) (base sample of 640 less 276 deals with missing data to construct variables less five singletons).

Dependent Variable	(1)	(2)	(3)	(4)
	<i>RivalRet</i>	<i>RivalRet</i>	$\Delta \text{Gross-Margin}$	$\Delta \text{Gross-Margin}$
JustBelowThreshold	0.005 (1.05)	0.002 (0.49)	-0.002 (-0.39)	-0.005 (-1.05)
Horizontal		-0.006* (-1.79)		-0.009*** (-3.18)
Horizontal $\times$ JustBelowThreshold		0.009* (1.78)		0.011** (2.82)
Constant	-0.064*** (-2.92)	-0.066*** (-2.91)	0.039 (0.77)	0.037 (0.74)
Observations	543	543	359	359
Adjusted R^2	0.115	0.115	0.066	0.066
Controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes

where $y_{i,[-1,1]}$ is the three-day market-adjusted return of an equal-weighted portfolio of horizontal rivals of the acquirer.

Table X, columns (1)–(2): The interaction **Horizontal** $\times$ **JustBelowThreshold** is positive and statistically significant, indicating that rivals react more favorably to horizontal deals that are just below the threshold than to other deals. This is consistent with such deals being perceived as more anticompetitive (or more likely to succeed without remedy) than comparable deals above the threshold.

6.2 Industry gross margins (Table X)

The paper also examines whether the post-merger environment shows higher markups/profitability. Using changes in industry-average gross margins as the outcome, **Table X, columns (3)–(4)** show a positive, significant interaction **Horizontal** $\times$ **JustBelowThreshold**. This aligns with a reduced-competition interpretation: industry profitability rises more following stealth horizontal mergers.

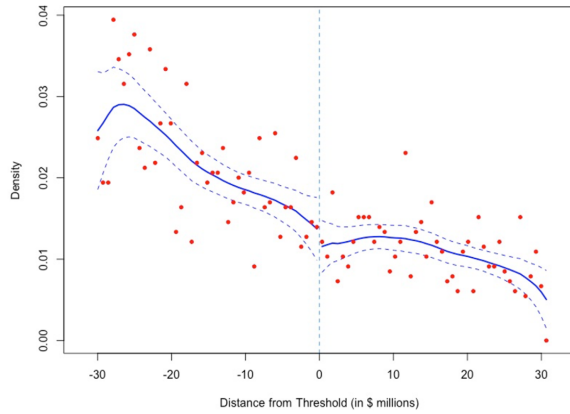


Figure 6. Falsification test: Density of M&As around FTC pre-merger review threshold involving public and private targets (using one-year ahead threshold level). This figure depicts the estimated kernel densities of the distance (in \$ millions) of deal values from the FTC pre-merger review threshold when we substitute the one-year-ahead threshold level for the current year's threshold level. Confidence bands (at the 95% level) are depicted by the thin lines. The circles, which describe scaled frequencies, are analogous to histogram bins. For ease of interpretation, we restrict our kernel density estimations to distances within $\pm \$30$ million of zero, where zero represents the FTC threshold. The estimation procedure follows the method in McCrary (2008). (Color figure can be viewed at wileyonlinelibrary.com)

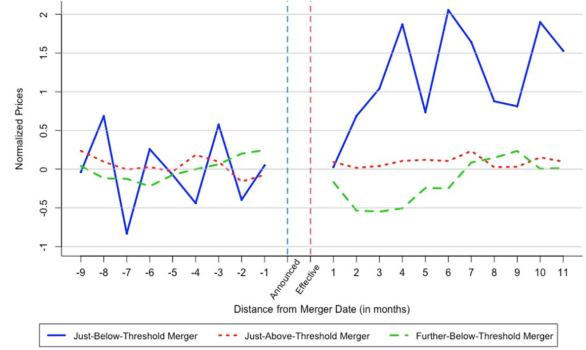


Figure 7. Product pricing before and after horizontal mergers. This figure plots monthly normalized average prices of approximately 1.9 million retail scanner observations collected before the announcement date and after the effective date for three horizontal mergers: one just below the threshold, one just above the threshold, and one further below the threshold. All data are for product sales of acquirers and rivals of the acquirer. Products are those that are common to the target and the acquirer (and rivals). Given these mergers occur at different points in time, the x-axis shows months relative to the announcement date (blue dotted line) and the effective date (red dotted line) for each merger. On the y-axis, normalized prices (i.e., mean = 0 and variance = 1) are displayed in the form of standard deviations, and thus observations on the graph can be interpreted as the number of standard deviations from the mean. Section III of the Internet Appendix provides details on the method used to define common products. We provide results from regressions using this sample in Table XI, which includes standard errors and tests of statistical significance. (Color figure can be viewed at wileyonlinelibrary.com)

6.3 Consumer prices using retail scanner data (Figure 7 and Table XI)

To directly probe consumer impacts, the paper uses a retail scanner panel for selected mergers and studies prices of products that are “common” to the target and acquirer (and rivals).

6.3.1 Difference-in-differences model (Table XI)

The main specification is:

$$y_{i,t} = \alpha + \beta_1 (\text{JustBelowThreshold}_{i,t} \times \text{Post}_t) + \beta_2 \text{JustBelowThreshold}_{i,t} + \beta_3 \text{Post}_t + \theta \text{Controls}_{i,t} + \varepsilon_{i,t}, \quad (5)$$

Table XI
Product-Level Prices for Below-Threshold M&As

This table presents results from difference-in-differences OLS regressions of normalized average weekly product prices on three mergers: one just below the threshold, one just above the threshold, and one further below the threshold. The dependent variable, *NormalizedPrice*, is a continuous variable that represents the normalized average weekly prices for common products of the acquirer and its rivals. The main variable of interest, *JustBelowThreshold* $\times$ *Post*, is an interaction term that takes the value of 1 if the observation belongs to the merger occurring just below the threshold and the week falls after the effective date of the merger, and 0 otherwise. All variables are defined in Appendix A. We vary the inclusion of week and geographic fixed effects across columns such that column (5) represents our fully specified model. For our week fixed effect, we count weeks relative to the effective date of the merger, since the three mergers occur in different years. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the product and week levels. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable	(1) <i>Normalized Price</i>	(2) <i>Normalized Price</i>	(3) <i>Normalized Price</i>	(4) <i>Normalized Price</i>	(5) <i>Normalized Price</i>
JustBelowThreshold	-0.000 (0.00)	0.002 (0.03)	-0.043 (-0.54)	0.168 (0.36)	0.181 (0.39)
Post	0.087 (1.29)	-0.200*** (-2.79)	-0.207*** (-2.85)		
JustBelowThreshold $\times$ Post	1.30* (1.93)	1.28* (1.98)	1.27** (2.00)	1.23** (2.21)	1.22** (2.30)
TimeTrend		0.001*** (3.58)	0.001*** (3.49)	0.064 (0.41)	0.088 (0.57)
Constant	-0.000 (0.00)	0.149* (1.71)	0.162* (1.93)	0.812 (0.44)	1.114 (0.62)
Observations	1,915,291	1,915,291	1,915,291	1,915,291	1,915,291
Adjusted R^2	0.004	0.010	0.137	0.044	0.173
Week fixed effects	No	No	No	Yes	Yes
Geographic fixed effects	No	No	Yes	No	Yes

where $y_{i,t}$ is the normalized average weekly price, and Post_t is an indicator for weeks after the merger becomes effective. The authors vary fixed effects (week and geography) across columns.

Figure 7 provides event-time intuition: prices are relatively stable pre-merger across the three example mergers, but the just-below (stealth) merger shows a sharp post-completion rise.

Table XI reports the DiD estimates. Across specifications (especially the fully specified column (5)), the interaction term is positive and statistically significant, with an economically large magnitude (about 1.2 standard deviations in normalized prices). This is consistent with the stealth merger reducing competitive pressure and allowing higher prices.

6.3.2 Within-merger “common vs uncommon” comparison (additional model)

The paper also discusses a within-merger design that compares common products to products not affected by the merger:

$$y_{i,t} = \alpha + \beta_1 (\text{CommonProduct}_{i,t} \times \text{Post}_t) + \beta_2 \text{CommonProduct}_{i,t} + \beta_3 \text{Post}_t + \theta \text{Controls}_{i,t} + \varepsilon_{i,t}, \quad (6)$$

where $\text{CommonProduct}_{i,t}$ indicates products that overlap with the target’s product set. (In the main paper, the detailed regression table for this specification is placed in the Internet Appendix; the core takeaway is that common products exhibit larger post-merger price increases.)

7 How to read the paper as an empirical strategy

7.1 Identification logic in one sentence

The key identifying “shock” is the discontinuous change in regulatory review requirements at the HSR threshold, combined with the fact that the threshold moves over time in a way plausibly exogenous to any one deal, enabling a standardized distance-to-threshold design.

7.2 What is descriptive vs. what is causal

- The **density discontinuity** is fundamentally descriptive evidence of bunching/manipulation (strong, but not a treatment-effect estimate).
- The **contracting regressions** (Tables IV–IX) are cross-sectional associations that help interpret *mechanisms* for avoidance.
- The **product-market outcome regressions** (Tables X–XI) aim to connect avoidance to competition outcomes. These rely on comparability of near-threshold deals and, in the retail-scanner setting, on a small set of well-measured cases with rich price data.

8 Main takeaway list (exam-style)

1. There is strong bunching just below the HSR threshold (Figures 2–3; Table III), robust to multiple falsifications (Figures 4–6).
2. Just-below deals disproportionately feature contract terms consistent with “value shifting” or “compensation off the headline price” (earnouts, insurance/indemnification structure, CEO economic ties) (Tables V–VIII).
3. Stealth behavior is especially prevalent in settings where anticompetitive incentives are strongest: horizontal, intrastate, and concentrated-industry mergers (Table IX).
4. Below-threshold horizontal deals are associated with outcomes consistent with reduced competition: positive rival announcement returns and higher industry gross margins (Table X), and higher consumer prices in retail scanner data (Figure 7; Table XI).

9 Conclusion

9.1 Summary

This paper concludes that **firms strategically structure M&A transactions to avoid antitrust scrutiny**, and that such avoidance is economically meaningful for product-market competition.

The empirical narrative is a tightly connected chain:

1. **Avoidance is visible in the deal-value distribution.** Around the U.S. pre-merger notification threshold (Hart–Scott–Rodino/FTC threshold), there is a sharp *bunching* of deals just below the filing cutoff and a corresponding *missing mass* just above it. Relative to a smooth counterfactual distribution, there are **about 40% more deals than expected** just below the threshold, and placebo/falsification thresholds do not show the same pattern.
2. **Avoidance is implemented through contract structure that makes deal values more malleable.** “Stealth acquisitions” are disproportionately deals (especially with **private targets**) whose contractual terms create *greater scope to negotiate and assign a lower reported transaction value*, including: earnouts/contingent payments, a higher reliance on cash/non-stock

structures, and legal/governance provisions that can compensate target-side stakeholders while keeping headline value under the threshold (e.g., extended D&O protections, higher deductibles for post-closing claims, and post-merger CEO economic ties).

3. **Stealth acquisitions are concentrated where anticompetitive incentives are strongest.** The bunching is stronger in settings consistent with greater incentives to coordinate and reduce competition: **horizontal** deals, especially in **concentrated industries** and for **intrastate** acquirer–target pairs.
4. **Product-market outcomes move in the direction of reduced competition.** Following stealth acquisitions, both acquirers and their **horizontal rivals** experience increases in equity values and margins, and product prices for common products rise in a retail-scanner case-study design. This pattern is consistent with **reduced competition (higher market power)** rather than pure efficiency improvements.

9.2 Key conclusions

- **Large discontinuity at the filing threshold.** Deal values cluster sharply below the threshold; the discontinuity is statistically strong and absent in falsification exercises. *Interpretation:* the threshold is a binding constraint that induces regulatory arbitrage.
- **Private targets and flexible contract terms drive stealth deals.** Stealth acquisitions are more likely for public acquirers buying private targets and for payment/contract terms that can shift consideration into hard-to-value components (e.g., earnouts) or reduce observable premiums. *Interpretation:* deal-value manipulation is feasible where valuation is inherently less verifiable.
- **The strongest bunching occurs in “high antitrust risk” settings.** Horizontal $\times$ (concentrated industry) and horizontal $\times$ (intrastate) interactions predict stealth deals. *Interpretation:* avoidance is not random; it is concentrated where scrutiny would matter most.
- **Rival returns and margins rise after stealth acquisitions.** Rivals’ announcement returns are positive and industry gross margins increase following just-below-threshold horizontal deals. *Interpretation:* rivals benefit when competition softens; this is a classic signature of anticompetitive consolidation.
- **Direct price evidence points the same way.** In Nielsen retail-scanner data for three horizontal mergers (one stealth just below the threshold, one just above, one further below), the stealth merger is associated with a **roughly 1.2 standard deviation** increase in prices for rivals’ products that overlap with the merged firms, while the non-stealth comparison mergers show no comparable price increase. *Interpretation:* in at least an observable product-market setting, stealth avoidance is linked to consumer-facing price increases.

9.3 Why a bright-line threshold creates “stealth” behavior

The core mechanism is **threshold-triggered regulatory arbitrage**.

Let T denote the filing threshold. Let V^{true} be the underlying economic value of the transaction, and let V^{rep} be the *reported/assignable* deal value used for threshold determination. Contractual features κ (earnouts, contingent payments, liability terms, side arrangements) create slack between the two:

$$V^{\text{rep}} = V^{\text{true}} - s(\kappa), \quad s(\kappa) \geq 0,$$

where $s(\kappa)$ captures the extent to which contract structure shifts value into components that are harder to pin down as “transaction value”.

Suppose the expected cost of review is Φ if the deal triggers filing:

$$\text{Cost of scrutiny} = \Phi \cdot \{V^{\text{rep}} \geq T\},$$

where Φ includes direct filing costs and, crucially, the risk of a costly “Second Request” investigation or a blocked deal.

Then the acquirer chooses κ (and bargaining terms with the target) to maximize

$$\max_{\kappa} \Pi(V^{\text{true}}, \kappa) - \Phi \cdot \{V^{\text{rep}} \geq T\}.$$

When Φ is large (e.g., in horizontal, concentrated markets), the incentive to choose κ that pushes V^{rep} just below T is strongest. This yields exactly the observed bunching.

9.4 Why positive rival returns and higher margins signal reduced competition

A key diagnostic comes from the idea that a merger can create:

- **Efficiency effects** (cost reductions, better products), which tend to *hurt* rivals through tougher competition; versus
- **Market-power effects** (less competition, higher prices), which tend to *benefit* rivals because the industry becomes more profitable.

Therefore, the finding that rivals’ equity values and industry gross margins rise after stealth horizontal mergers is naturally read as evidence that stealth acquisitions are disproportionately those with **market-power** implications.

9.5 Policy implication (what the conclusion implies for antitrust design)

The paper’s conclusion is not merely that many deals fall below a threshold, but that **thresholds can be manipulated through contract design**. This implies:

1. bright-line deal-value thresholds may under-screen anticompetitive consolidation when parties can repackage consideration;
2. enforcement resources may be better targeted using *risk signals* beyond headline value (e.g., horizontal overlap, market concentration, geography, and contract features that enable value assignability);

3. corporate finance contracting is part of the antitrust problem: **financial and governance terms can serve as tools of regulatory avoidance.**

9.6 Takeaway

Kepler, Naiker, Stewart (2023) show that firms bunch M&A deal values just below the antitrust filing threshold—especially in horizontal, concentrated settings and in private-target deals with flexible contracting terms—and that these “stealth acquisitions” are followed by higher rival valuations, higher industry margins, and case-study evidence of higher product prices, consistent with regulatory avoidance that reduces product-market competition and harms consumers.